MIL-D-22994(SHIPS)
1 September 1961

MILITARY SPECIFICATION

DOME, SONAR, RUBBER, 60-INCH, TYPE CW-177D/U

1. SCOPE

1.1 This specification covers the design and construction of a non-magnetic, 60-inch rubber dome intended for installation and use aboard Naval ships.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-15024 - Plates, Identification, Information for Identification of Electrical, Electronic, and Mechanical Equipment.
MIL-P-15931 - Paint, Antifouling.

STANDARDS

FEDERAL

FED-STD-601 - Rubber; Sampling and Testing.

MILITARY

MIL-STD-278 - Welding and Allied Processes for Machinery for the United States Navy.

FSC 5845

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:04Z
Check the source to verify that this is the current version before use.

DRAWINGS

BUREAU OF SHIPS
MICROFILM REEL NO. 26586 - Dome, Sonar, Rubber, Type CW-177D/U.

(Copies of specifications, standards, drawings, and publications required by the contractor in connection with procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N.Y.).

3. REQUIREMENTS

3.1 Materials.- Unless otherwise specified herein, materials used in the construction of sonar dome CW-177D/U shall be as shown on microfilm reel No. 26586.

3.1.1 Nonmagnetic material.- All material entering into the construction of the complete dome shall be nonmagnetic. Material with a permeability of less than 1.08 before fabrication and no greater than 2.0 on the finished product, shall be considered nonmagnetic. Wherever possible, preference shall be given to nonmagnetic material having the lowest electrical conductivity.

3.2 General requirements.- Unless otherwise specified herein, the design, contours, dimensions and requirements for the sonar dome CW-177D/U shall be as shown on microfilm reel No. 26586.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:04Z
Check the source to verify that this is the current version before use.

3.3 Design.- The dome shall be a modified Navy type Sonar Rubber Dome CW-177C/U, and shall be interchangeable with the CW-177C/U. It shall be equal to or better than the CW-177C/U in its capability to withstand shock, vibration, and stress encountered in service use. In addition, the dome shall be reinforced to provide resistance to side bending under a high impact load normally encountered in heavy seas. Such reinforcements shall consist basically of cross braces in the fore and aft sections, truss work around the lower section, and gussets on the vertical side members. The dome shall be so designed and constructed that the rubber portion will not part from the corrosion-resisting steel portion at the seam, under any condition likely to be encountered in service use, and shall be capable of withstanding a 34-foot static head of water in order that transducers can be changed while the ship is water borne.

3.4 Sound transparency.- The dome shall be designed for use with the mine detecting sets AN/UQS-1b and 1d and shall have a sound transparency to conform to the AN/UQS-1b, 1d underwater field of coverage of 360 degrees in azimuth, and from 0 to -50 degrees in the vertical plane. Some deterioration of performance is acceptable in the aft portion of the dome between 120 and 240 degrees, as measured from the acoustic centerline of the transducer assembly.

3.5 Dome mounting flange.- The surfaces of the dome mounting flange shall be within plus or minus 1/32 inch of their plane.

3.6 Interchangeability.- All CW-177D/U domes produced under a contract shall be interchangeable.

3.7 Welding.- Welding shall conform to Standard MIL-STD-278.

3.8 Antifouling paint.- Antifouling paint shall be in accordance with Specification MIL-P-15931.

3.9 Identification plates.- Corrosion-resisting non-magnetic steel identification plates bearing markings and assigned designation shall be furnished in accordance with type A, B or C of Specification MIL-P-15024.

3.10 Deviations.- Minor deviations from microfilm reel No. 26586 necessitated by shop practices, and deemed essential to ensure superior workmanship will be permitted, subjected to the approval of the bureau or agency concerned as having no adverse effects upon the required material performance. Any such minor changes which are approved by the bureau or agency concerned shall be incorporated in the manufacturing drawings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:04Z
Check the source to verify that this is the current version before use.

3.11 Workmanship. - Workmanship shall be of high quality.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Testing of rubber. - Rubber shall be tested in accordance with Standard FED-STD-601.

4.3 At the discretion of the bureau or agency concerned, the dome specified herein shall be shipped, to a ship or station designated by the bureau or agency concerned, for a shipboard evaluation test to determine conformance with this specification. Prior to submission for tests, the contractor shall have conducted sufficient tests, to determine that the equipment conforms to this specification.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation. -

5.1.1 Sonar domes. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:04Z
Check the source to verify that this is the current version before use.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, packaging, packing, and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made (see 6.1).

5.2.1.1 Preservation, packaging, packing, and marking.- Sonar domes shall be individually preserved and packaged level A or C; packed level A or B and marked in accordance with Specification MIL-E-17555.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preservation, packaging, packing, and marking requirements other than those required by 5.1 (see 5.2)

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Ships
(Project 5845-N197Sh)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:04Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Instructions

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D. C.

Specification

Organization	City	State
Contract No.		
Quantity of Items Procured		Dollar Amount $

Material procured under a direct Government contract ☐ or a subcontract ☐

1. Has any part of the specification created problems or required interpretation in procurement?
   a. Give paragraph number and wording.

   b. Recommendations for correcting the deficiencies.

2. Comment on any specification requirement considered too rigid.

3. Is the specification restrictive? ☐ Yes ☐ No If the answer is "Yes", in what way?

4. Remarks (Attach any pertinent data which may be of use in improving this specification.) Place this form and papers in an envelope and send to the Bureau.

Submitted by (Print name and activity)	Date

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:04Z
Check the source to verify that this is the current version before use.